Homework 6

Solutions

Exercise 6.1. Suppose that $X_1, X_2, \dots, X_n \overset{i.i.d.}{\sim} \mathrm{Ber}(p)$ and let $\hat{p} = \frac{1}{n}\sum_{i=1}^n X_i$.

1. What is the bias, $\mathrm{bias}(\hat{p}, p)$?
2. What is the variance of $\hat{p}$?
3. What is the logarithm of the probability mass function, $\log f_{X_1;p}(k)$?
   Note: only compute $\log f_{X_1;p}(k)$ *where* $f_{X_1;p}(k) \neq 0$
4. Use this to compute the Fisher information, $I(p) = \mathbb{E}\left[\left(\frac{\partial}{\partial p}\log f_{X_1,p}(X_1)\right)^2\right]$.
5. Show that $\hat{p}$ attains the Cramér-Rao lower bound for unbiased estimators.
6. Show that $\hat{p}$ maximizes the log-likelihood function $\ell(p)$, and hence the likelihood function $L(p)$ as well.

Sol. First, the bias is given by

$$\mathrm{bias}(\hat{p}, p) = \mathbb{E}[\hat{p}] - p = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] - p = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] - p = \frac{np}{n} - p = \boxed{0}$$

using the linearity of expectation and the fact that $\mathbb{E}[\mathrm{Ber}(p)] = p$. Since the samples are independent, we can similarly compute the variance as

$$\mathrm{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i) = \frac{np(1-p)}{n^2} = \boxed{\frac{p(1-p)}{n}}$$

using that $\mathrm{Var}(\mathrm{Ber}(p)) = p(1-p)$. As a side-note, since the estimator $\hat{p}$ is unbiased, this variance is also the mean-squared error, $\mathrm{MSE}(\hat{p}, p)$. Now, we can compute directly that

$$\log f_{X_1;p}(k) = \boxed{\begin{cases} \log p & \text{if } k = 1 \\ \log(1-p) & \text{if } k = 0 \end{cases}}$$

and is not defined otherwise. Hence,

$$\frac{\partial}{\partial p}\log f_{X_1;p}(k) = \begin{cases} \frac{1}{p} & \text{if } k = 1 \\ \frac{-1}{1-p} & \text{if } k = 0 \end{cases}$$

and so the Fisher-information is given by

$$I(p) = \mathbb{E}\left[\left(\frac{\partial}{\partial p}\log f_{X_1,p}(X_1)\right)^2\right] = \frac{1}{p^2}\mathbb{P}(X_1 = 1) + \frac{1}{(1-p)^2}\mathbb{P}(X_1 = 0) = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \boxed{\frac{1}{p(1-p)}}.$$

So, the Cramér-Rao lower bound reads, for any unbiased estimator T,

$$\mathrm{Var}(T) \geq \frac{(1+0)^2}{nI(p)} = \frac{1}{\frac{n}{p(1-p)}} = \frac{p(1-p)}{n}.$$

The right hand side is exactly $\mathrm{Var}(\hat{p})$, so $\hat{p}$ attains the Cramér-Rao lower bound and is an optimal unbiased estimator of p.

As we computed in class, the likelihood function can be expressed as

$$L(p) = \prod_{i=1}^{n} f_{X_i;p}(X_i) = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} = p^{\sum_{i=1}^n X_i}(1-p)^{n-\sum_{i=1}^n X_i}$$

so the log-likelihood function (which has the same extremal points as $L(p)$) is given by

$$\ell(p) = \left(\sum_{i=1}^{n} X_i\right)\log p + \left(n - \sum_{i=1}^{n} X_i\right)\log(1-p).$$

Differentiating in p and setting the result equal to zero to find extrema,

$$0 = \frac{\partial}{\partial p}\ell(p) = \frac{1}{p}\left(\sum_{i=1}^{n} X_i\right) - \frac{1}{1-p}\left(n - \sum_{i=1}^{n} X_i\right) \iff \frac{1}{p}\left(\sum_{i=1}^{n} X_i\right) = \frac{n}{1-p} - \frac{n}{1-p}\sum_{i=1}^{n} X_i$$

which implies

$$\left(\frac{1}{p} - \frac{1}{1-p}\right)\sum_{i=1}^{n} X_i = \frac{n}{1-p} \iff \frac{1}{p(1-p)}\sum_{i=1}^{n} X_i = \frac{n}{1-p} \iff p = \frac{1}{n}\sum_{i=1}^{n} X_i = \hat{p}.$$

To see that this is indeed a maximum, notice that

$$\frac{\partial^2}{\partial p^2}\ell(p) = \frac{-1}{p^2}\sum_{i=1}^{n} X_i - \frac{1}{(1-p)^2}\left(n - \sum_{i=1}^{n} X_i\right) \leq 0$$

since $0 \leq X_i \leq 1 \implies 0 \leq \sum_{i=1}^n X_i \leq n$. Thus, $\ell(p)$ (and hence $L(p)$ as well) attains a maximum at $\hat{p}$; $\hat{p}$ is a maximum-likelihood estimator.

Exercise 6.2. Suppose that $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exp}(\lambda)$, and let $\hat{\lambda} = \frac{1}{n}\sum_{i=1}^n X_i$.

1. What is the bias with respect to $\frac{1}{\lambda}$, bias $\left(\hat{\lambda}, \frac{1}{\lambda}\right)$?

2. What is the variance of $\hat{\lambda}$?

3. Let $\theta := \frac{1}{\lambda}$. What is the logarithm of the probability density function, $\log f_{X_1;\theta}(x)$?

4. Use this to compute the Fisher information,

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f_{X_1,\theta}(X_1)\right)^2\right].$$

   Note: write everything in terms of $\theta = \frac{1}{\lambda}$ before differentiating, because we are differentiating with respect to the (natural) parameter $\theta = \frac{1}{\lambda}$ instead of λ.

5. Show that $\hat{\lambda}$ attains the Cramér-Rao lower bound for unbiased estimators.

6. Show that $\hat{\lambda}$ maximizes the log-likelihood function $\ell(\lambda)$, and hence the likelihood function $L(\lambda)$ as well.
   Note that here you will differentiate with respect to λ, not θ

Sol. First, the bias is given by

$$\text{bias}\left(\hat{\lambda}, \frac{1}{\lambda}\right) = \mathbb{E}[\hat{\lambda}] - \frac{1}{\lambda} = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] - \frac{1}{\lambda} = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] - \frac{1}{\lambda} = \frac{n}{n\lambda} - \frac{1}{\lambda} = \boxed{0}$$

using the linearity of expectation and the fact that $\mathbb{E}[\text{Exp}(\lambda)] = \frac{1}{\lambda}$. Since the samples are independent, we can similarly compute the variance as

$$\text{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(X_i) = \frac{n}{n^2\lambda^2} = \boxed{\frac{1}{n\lambda^2}}$$

using that $\text{Var}\left(\text{Exp}(\lambda)\right) = \frac{1}{\lambda^2}$. As a side-note, since the estimator $\hat{\lambda}$ is unbiased, this variance is also the mean-squared error, $\text{MSE}\left(\hat{\lambda}, \frac{1}{\lambda}\right)$. Now, we can compute directly that

$$\log f_{X_1;\lambda}(x) = \log\left(\lambda e^{-\lambda x}\right) = \log\lambda + \log\left(e^{-\lambda x}\right) = \log\lambda - \lambda x$$

and is not defined otherwise. Hence, writing this in terms of the natural parameter $\theta = \frac{1}{\lambda}$,

$$\log f_{X_1;\theta}(x) = \log\frac{1}{\theta} - \frac{x}{\theta} = \boxed{-\log\theta - \frac{x}{\theta}}$$

and is not defined otherwise. Differentiating in θ,

$$\frac{\partial}{\partial\theta}\log f_{X_1;\theta}(x) = \frac{-1}{\theta} + \frac{x}{\theta^2}$$

and so the Fisher-information is given by

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f_{X_1,\theta}(X_1)\right)^2\right] = \mathbb{E}\left[\left(\frac{X_1-\theta}{\theta^2}\right)^2\right] = \frac{1}{\theta^4}\mathbb{E}\left[(X_1-\theta)^2\right] = \frac{\text{Var}(X_1)}{\theta^4} = \frac{\theta^2}{\theta^4} = \boxed{\frac{1}{\theta^2} = \lambda^2}$$

where we have used that $\mathbb{E}[X_1] = \frac{1}{\lambda} = \theta$ and $\text{Var}(X_1) = \frac{1}{\lambda^2} = \theta^2$. So, the Cramér-Rao lower bound reads, for any unbiased estimator T,

$$\text{Var}(T) \geq \frac{(1+0)^2}{nI(\theta)} = \frac{1}{\frac{n}{\theta^2}} = \frac{1}{n\lambda^2}.$$

The right hand side is exactly $\text{Var}\left(\hat{\lambda}\right)$, so $\hat{\lambda}$ attains the Cramér-Rao lower bound and is an optimal unbiased estimator of $\theta = \frac{1}{\lambda}$.

Now, the likelihood function can be expressed as

$$L(\lambda) = \prod_{i=1}^{n} f_{X_i;\lambda}(X_i) = \prod_{i=1}^{n}\lambda e^{-\lambda X_i} = \lambda^n e^{-\lambda\sum_{i=1}^{n} X_i}$$

where it is nonzero, so the log-likelihood function (which has the same extremal points as $L(\lambda)$) is given by

$$\ell(\lambda) = n\log\lambda - \lambda\sum_{i=1}^{n} X_i.$$

Differentiating in λ and setting the result equal to zero to find extrema,

$$0 = \frac{\partial}{\partial\lambda}\ell(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} X_i \implies \lambda = \frac{1}{n}\sum_{i=1}^{n} X_i = \hat{\lambda}.$$

To see that this is indeed a maximum, notice that

$$\frac{\partial^2}{\partial\lambda^2}\ell(\lambda) = \frac{-n}{\lambda^2} < 0.$$

Thus, $\ell(\lambda)$ (and hence $L(\lambda)$ as well) attains a maximum at $\hat{\lambda}$; $\hat{\lambda}$ is a maximum-likelihood estimator.

Exercise 6.3. Suppose that $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \mathcal{N}\left(0, \sigma^2\right)$, and let $\hat{\theta} = \frac{1}{n}\sum_{i=1}^n X_i^2$ be our variance estimator in the situation where we know the mean (here zero).

1. What is the bias with respect to the (natural) parameter $\theta = \sigma^2$, bias $\left(\hat{\theta}, \theta\right)$?
2. What is the variance of $\hat{\theta}$?
3. Let $\theta := \sigma^2$. What is the logarithm of the probability density function, $\log f_{X_1;\theta}(x)$?
4. Use this to compute the Fisher information,

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f_{X_1,\theta}(X_1)\right)^2\right].$$

   Note: write everything in terms of $\theta = \sigma^2$ before differentiating, because we are differentiating with respect to the (natural) parameter $\theta = \frac{1}{\lambda}$ instead of λ. You may want to use that for a $Z \sim \mathcal{N}(0,1)$ random variable, $\mathbb{E}\left[Z^4\right] = 3$.

5. Show that $\hat{\theta}$ attains the Cramér-Rao lower bound for unbiased estimators.

Sol. First, the bias is given by

$$\text{bias}\left(\hat{\theta}, \theta\right) = \mathbb{E}[\hat{\theta}] - \sigma^2 = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i^2\right] - \sigma^2 = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i^2] - \sigma^2 = \frac{1}{n}\sum_{i=1}^n \text{Var}(X_i)] - \sigma^2 = \frac{n\sigma^2}{n} - \sigma^2 = \boxed{0},$$

where we have used the linearity of expectation and replaced $\mathbb{E}[X_i^2]$ with $\text{Var}(X_i)$ since $\mathbb{E}[X_i] = 0$. Since the samples are independent and identically distributed, we can compute the variance as

$$\text{Var}\left(\sum_{i=1}^n X_i^2\right) = \frac{1}{n^2}\sum_{i=1}^n \text{Var}(X_i^2) = \frac{n\text{Var}(X_1^2)}{n^2} = \frac{\text{Var}(X_1^2)}{n}.$$

Now, to find $\text{Var}(X_1^2)$, notice that

$$\text{Var}(X_1^2) = \mathbb{E}\left[\left(X_1^2\right)^2\right] - \left(\mathbb{E}\left[X_1^2\right]\right)^2 = \mathbb{E}\left[X_1^4\right] - \sigma^4$$

where we have used that $\mathbb{E}\left[X_1^2\right] = \text{Var}(X_1) = \sigma^2$. Now, $X_1 \sim \mathcal{N}(0, \sigma^2)$, so $\frac{X_1}{\sigma} \sim Z = \mathcal{N}(0,1)$ and thus

$$\frac{1}{\sigma^4}\mathbb{E}[X_1^4] = \mathbb{E}\left[\left(\frac{X_1}{\sigma}\right)^4\right] = \mathbb{E}[Z^4] = 3$$

where we have used that $\mathbb{E}[Z^4] = 3$. So, $\mathbb{E}[X_1^4] = 3\sigma^4$, and thus

$$\text{Var}(X_1^2) = \mathbb{E}\left[X_1^4\right] - \sigma^4 = 3\sigma^4 - \sigma^4 = 2\sigma^4 = 2\theta^2.$$

So,

$$\text{Var}(\hat{\theta}) = \boxed{\frac{2\theta^2}{n}}.$$

As a side-note, since the estimator $\hat{\theta}$ is unbiased, this variance is also the mean-squared error, $\text{MSE}\left(\hat{\theta}, \theta\right) = \frac{2\theta^2}{n}$. Now, we can compute directly that

$$\log f_{X_1;\theta}(x) = \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{x^2}{2\sigma^2}}\right) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{x^2}{2\sigma^2} = -\frac{1}{2}\log(2\pi\theta) - \frac{x^2}{2\theta}.$$

Differentiating in θ,

$$\frac{\partial}{\partial\theta}\log f_{X_1;\theta}(x) = \frac{-1}{2(2\pi\theta)}(2\pi) + \frac{x^2}{2\theta^2} = \frac{x^2-\theta}{2\theta^2}.$$

So, the Fisher-information is given by

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f_{X_1,\theta}(X_1)\right)^2\right] = \mathbb{E}\left[\left(\frac{X_1^2-\theta}{2\theta^2}\right)^2\right] = \frac{1}{2\theta^4}\mathbb{E}\left[\left(X_1^2-\theta\right)^2\right] = \frac{\mathrm{Var}(X_1^2)}{2\theta^4} = \frac{\theta^2}{2\theta^4} = \boxed{\frac{1}{2\theta^2} = \frac{1}{2\sigma^4}}$$

where we have used that $\mathbb{E}[X_1^2] = \sigma^2 = \theta$ and $\mathrm{Var}(X_1^2) = 2\theta^2$. So, the Cramér-Rao lower bound reads, for any unbiased estimator T,

$$\mathrm{Var}(T) \geq \frac{(1+0)^2}{nI(\theta)} = \frac{1}{\frac{n}{2\theta^2}} = \frac{2\theta^2}{n} = \frac{2\sigma^4}{n}.$$

The right hand side is exactly $\mathrm{Var}\left(\hat{\theta}\right)$, so $\hat{\lambda}$ attains the Cramér-Rao lower bound and is an optimal unbiased estimator of $\theta = \sigma^2$.